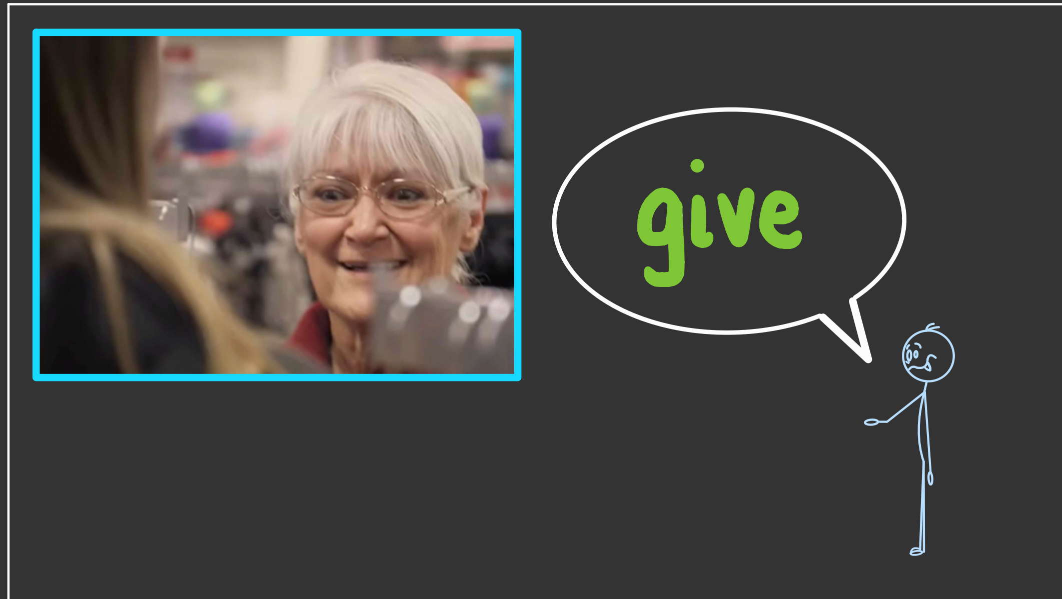
function expressions

CSCI 134

previously on csci 134...

we observed that verbs and functions typically require arguments to have a "complete" meaning



but they still mean something

on their own

give

what? to whom?

people understand giving as an abstract concept, even without specifying the arguments

pow

base? exp?

similarly, python understands pow as an abstract concept, even without specifying the arguments

```
In [1]: pow(6, 2)
Out[1]: 36

In [2]: pow
Out[2]: <function pow(base, exp)>
```

similarly, python understands pow as an abstract concept, even without specifying the arguments

```
In [1]: abs(-7.2)
Out[1]: 7.2

In [2]: abs
Out[2]: <function abs(x)>

In [3]: len("cookie")
Out[3]: 6

In [4]: len
Out[4]: <function len(obj)>
```

this is also true for the other **functions** we have encountered

```
In [1]: type(abs)
Out[1]: builtin_function_or_method

In [2]: type(max)
Out[2]: builtin_function_or_method

In [3]: type(pow)
Out[3]: builtin_function_or_method

In [4]: type(len)
Out[4]: builtin_function_or_method
```

a function is just another type of expression

expressions so far

type	example	value
int	7	7
float	3.14	3.14
str	"williams"	"williams"
bool	3 > 4	False
function	abs	`<function abs(x)>`